

Lecture Unit 3

MATH 5394 **Markov Chains and Algorithmic Applications**

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Fall Semester, 2026

September 24, 2026

Computing Framework and Software Tools

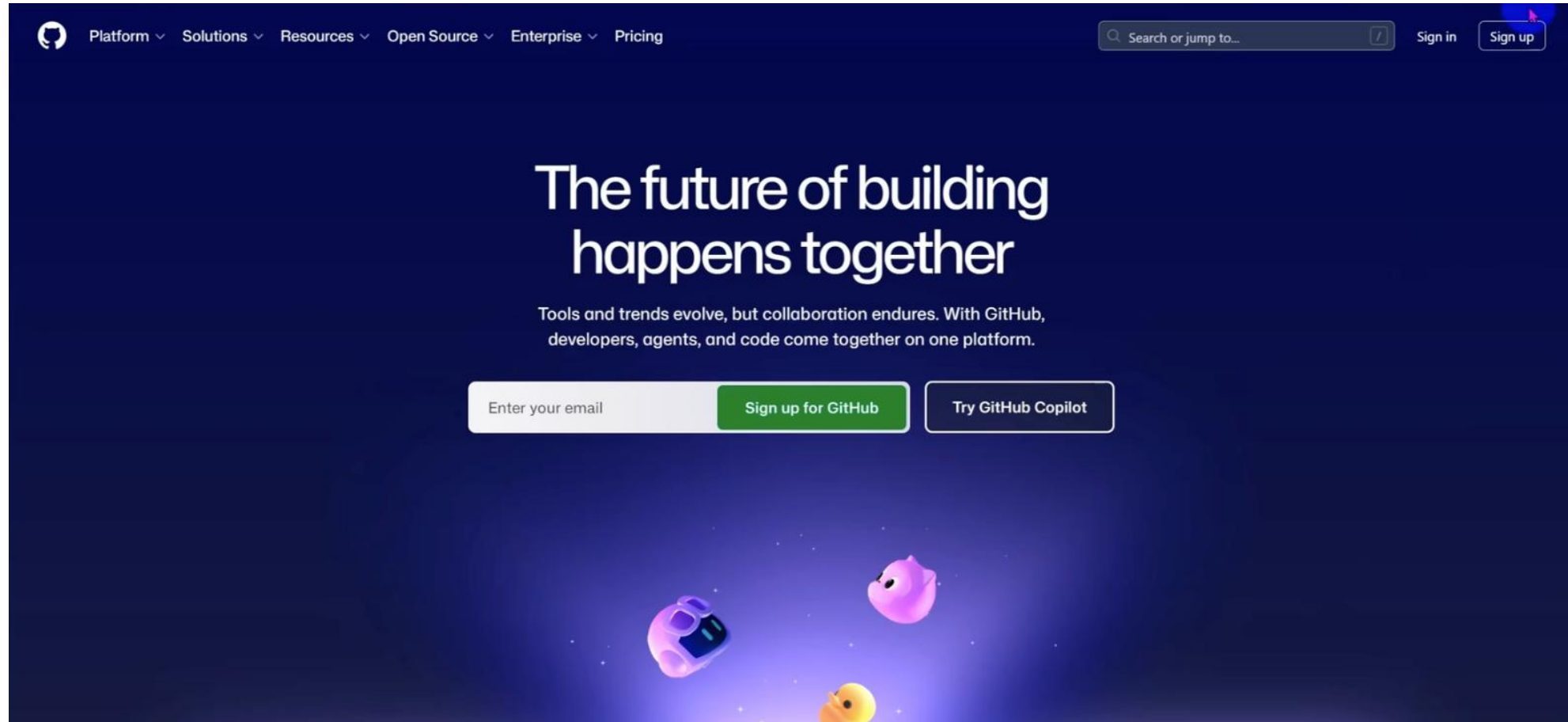
GitHub

MMAT 5394

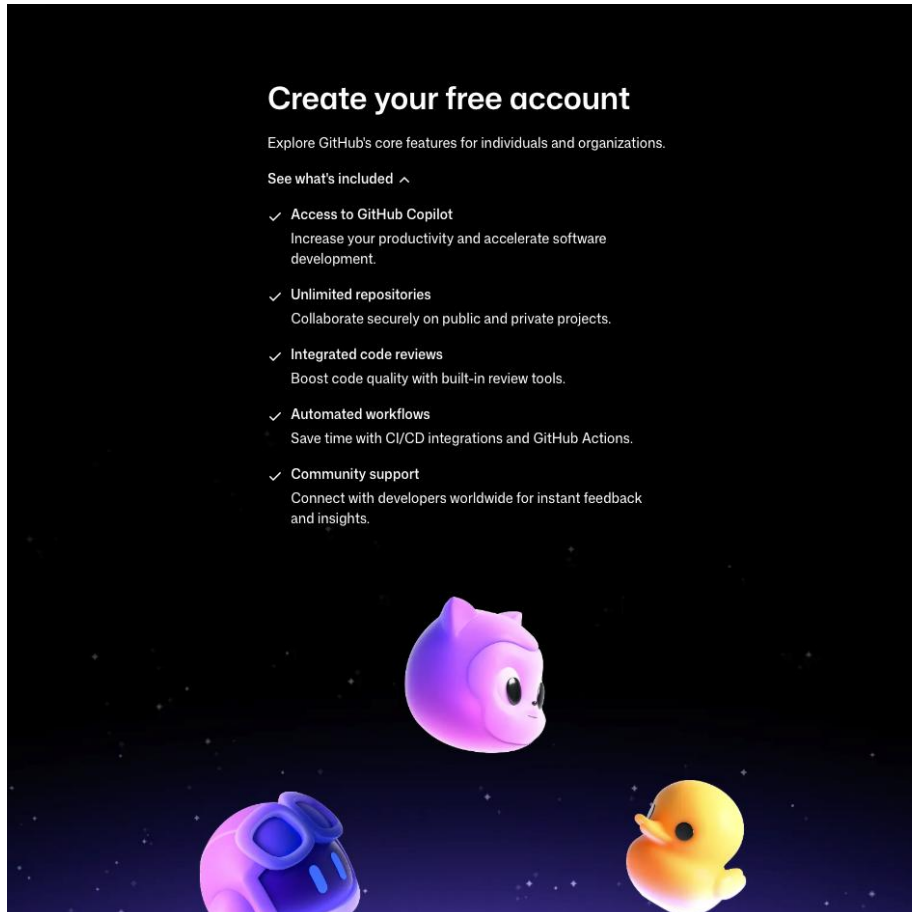
Zishang Li

<https://github.com/>

Sign in or sign up



Continue with Google or Apple or Create account by email



Already have an account? [Sign in](#)

Sign up for GitHub

Continue with Google

Continue with Apple

or

Email

Email

Password

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username

Username

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region

Hong Kong

For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

GitHub Copilot

☒ Sign up for Copilot Free for coding assistance on the web, in the IDE, and in the CLI. [See terms](#)

Email preferences

☐ Receive occasional product updates and announcements

Create account >

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll

Repository(Repo) is like an online project folder stored on GitHub.

Repository = project folder + version history.
You can see what changed, when, and by whom.

Public or private.

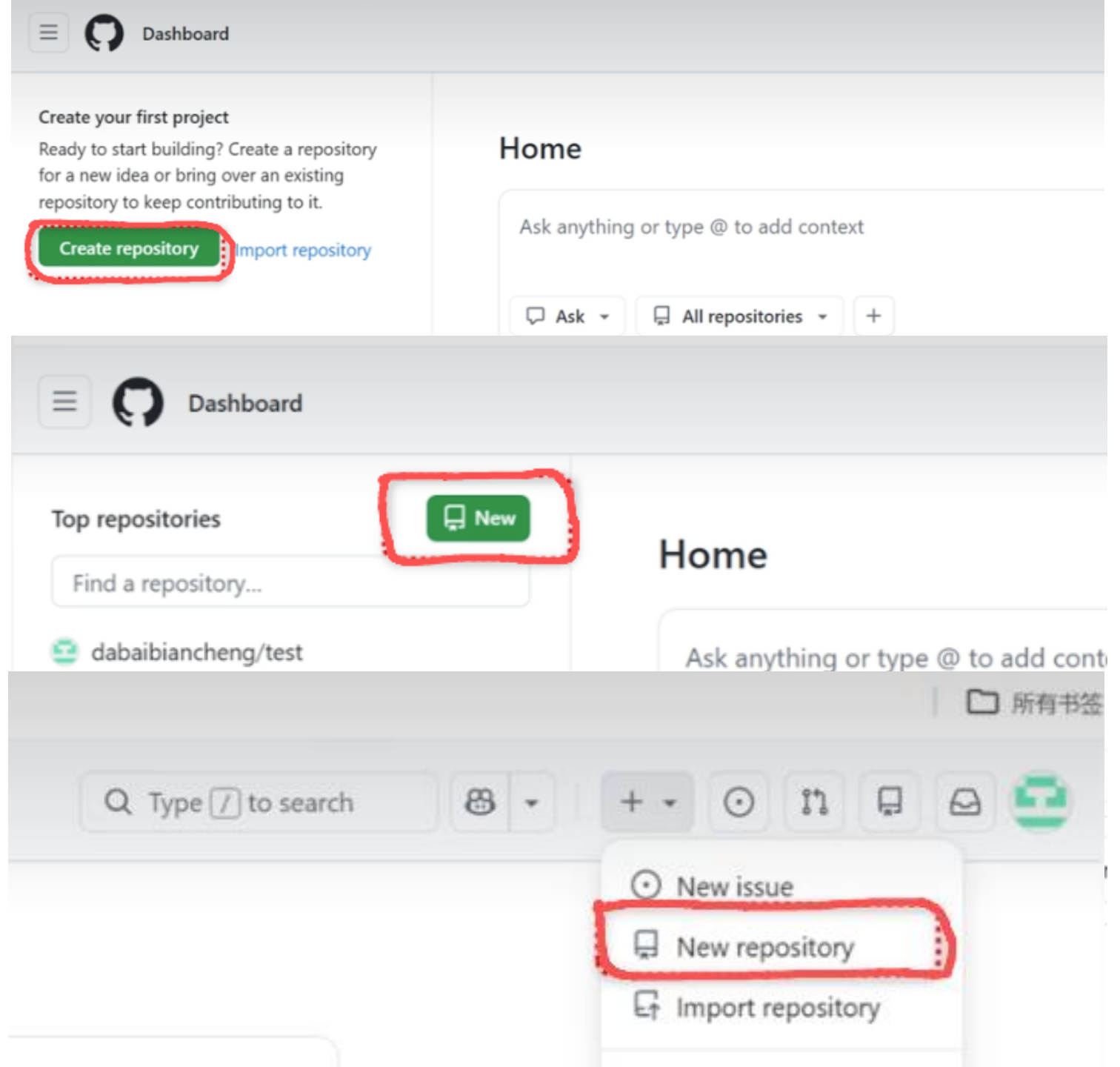
A public repo can be viewed by anyone. A private repo can only be seen by invited people.

Collaboration tools.

People can suggest changes, report problems, and work together using features like branches, pull requests, and issues.

README file.

A README.md file usually explains what the project is and how to use it.



- **Owner**
Usually, this is your personal GitHub account.
- **Repository name**
Enter a name for your project. This is required. For example: python-homework.
- **Description** (optional)
- **Public / Private**
 - **Public:** Anyone can see the repository.
 - **Private:** Only you and invited people can see it.
- **Initialize this repository with:**
You can choose to add some starter files:
 - **Add a README file**
Creates a README.md file. It is a good place to explain what the project is and how to use it.
 - **Add .gitignore**
Lets you choose a template to ignore files you do not want to upload, such as temporary or system files.(Keep the default options)
 - **Choose a license**
Adds a license file to tell others how they may use your code.(Keep the default options)
- **Create repository**
After you fill in the name and choose the settings, click this button. GitHub will create your new repository.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

Repository name *

 zsli-cuhk ▾

 /

Great repository names are short and memorable. How about [congenial-octo-goggles](#)?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

Create repository

Code – The main page. Shows all files and folders in the repository.

...

Settings – Only the owner can see this. Used to rename the repo, change visibility, manage branches, add collaborators, or delete the repo.

Main Code Tab

Branch button – Switch between branches or tags.

File list – Shows all files and folders. Click a file to view it.

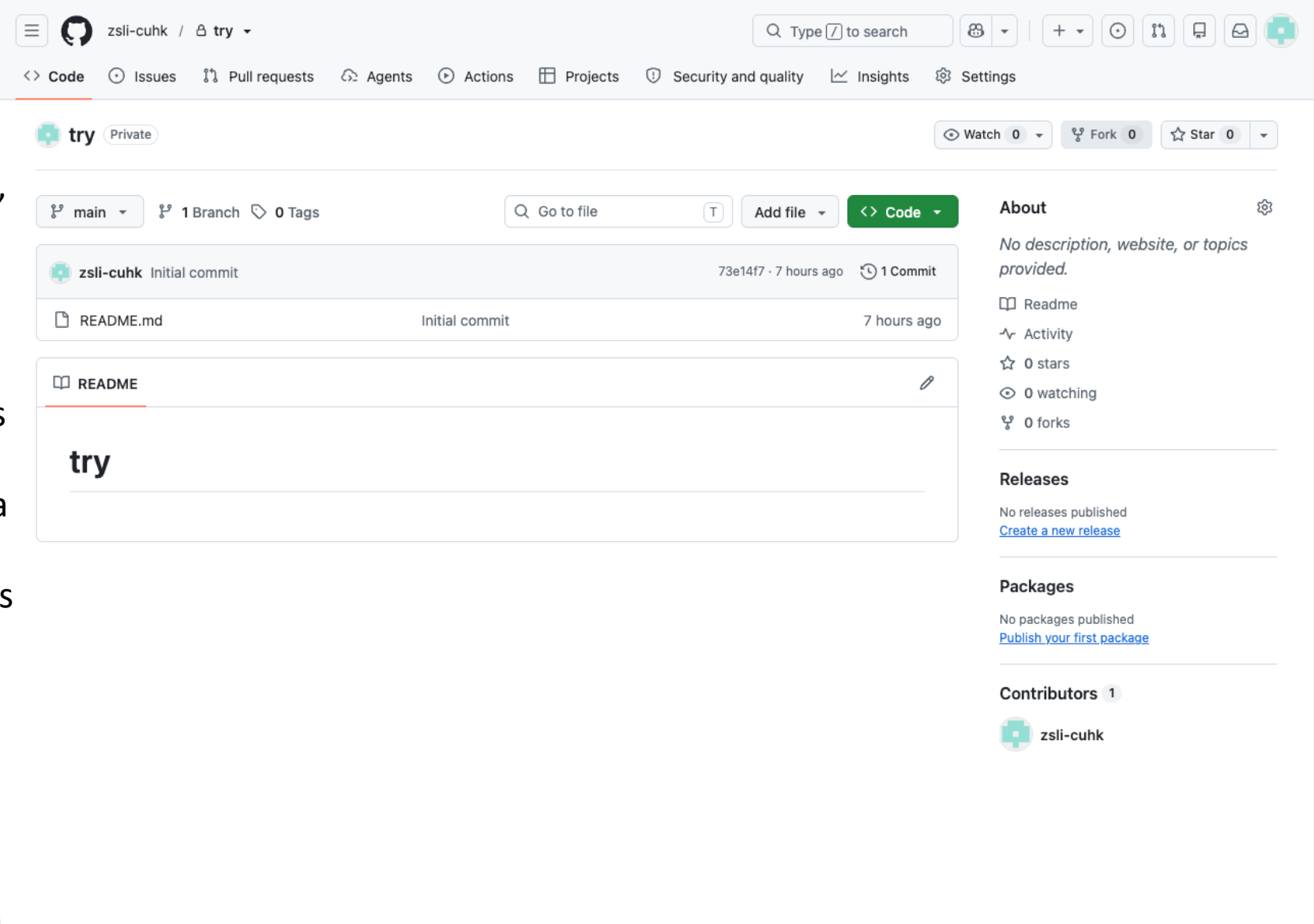
README file – If you added one, it appears below the file list. It usually explains the project.

Go to file – Quickly search for a file.

Add file – Create a new file or upload files from your computer.

Code button – Gives you options to clone the repo using HTTPS, SSH, or GitHub CLI, or download it as a ZIP file.

Commit message – Shows the latest change and when it was made.



- Create new files
- Upload files

zsli-cuhk / try

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

try Private Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file

Add file

- + Create new file
- Upload files

zsli-cuhk Update helloworld.py b32b

README.md	Initial commit	9 hours ago
build_medium.py	upload build medium file	2 hours ago
helloworld.py	Update helloworld.py	2 hours ago

README

try

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

zsli-cuhk

Languages

Python 100%

Suggested workflows

<https://github.com/zsli-cuhk/try/new/main>

Search MMAT5394

Read README.md

This screenshot shows the GitHub search interface. The search bar at the top left contains the text 'MMAT5394' and is highlighted with a red box. Below the search bar, the left sidebar shows the 'Filter by' section with 'Repositories' selected, showing 1 result. The main content area displays '1 result (85 ms)' and a single repository result for 'zsli-cuhk/MMAT5394', which is highlighted with a red box. The repository details show it is a Python project with 0 stars and was updated 2 minutes ago. The right sidebar shows the repository's README content, which includes a 'Sponsor open source projects you depend on' section and a 'ProTip!' about using the forward slash key to refine search queries.

Filter by

- <> Code 1
- Repositories 1
- Issues 0
- Pull requests 0
- Discussions 0
- Users 0
- More

Advanced

- + Owner
- + Size
- + Number of followers
- + Number of forks
- + Number of stars
- + Date created
- + Date pushed
- + Topic
- + License

1 result (85 ms) Sort by: Best match Save

zsli-cuhk/MMAT5394 ☆ Star

Python · ☆ 0 · Updated 2 minutes ago

Sponsor open source projects you depend on

Contributors are working behind the scenes to make open source better for everyone—give them the help and recognition they deserve.

[Explore sponsorable projects →](#)

How can we improve search?

[Give feedback](#)

ProTip! Press the `/` key to activate the search input again and adjust your query.

This screenshot shows the GitHub repository page for 'zsli-cuhk/MMAT5394'. The search bar at the top right contains the text 'Type / to search' and is highlighted with a red box. The repository details show it is a Python project with 0 stars and was updated 2 minutes ago. The main content area displays the repository's README content, which includes a 'Sponsor open source projects you depend on' section and a 'ProTip!' about using the forward slash key to refine search queries. The right sidebar shows the repository's README content, which includes a 'Sponsor open source projects you depend on' section and a 'ProTip!' about using the forward slash key to refine search queries.

zsli-cuhk / try

Search Type / to search

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

try Private Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file Code

zsli-cuhk Update helloworld.py b32bd56 · 2 hours ago 4 Commits

README.md	Initial commit	9 hours ago
build_medium.py	upload build medium file	2 hours ago
helloworld.py	Update helloworld.py	2 hours ago

MMAT5394

About

No description, website, or topics provided.

- Readme
- Activity
- 0 stars
- 0 watching
- 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

zsli-cuhk

Languages

Python 100%

Suggested workflows

For your submission:

- Students can click **Fork** to copy the public repo.
- Upload your files in their fork using **Add file**.
- Open a **Pull request** to submit your work.

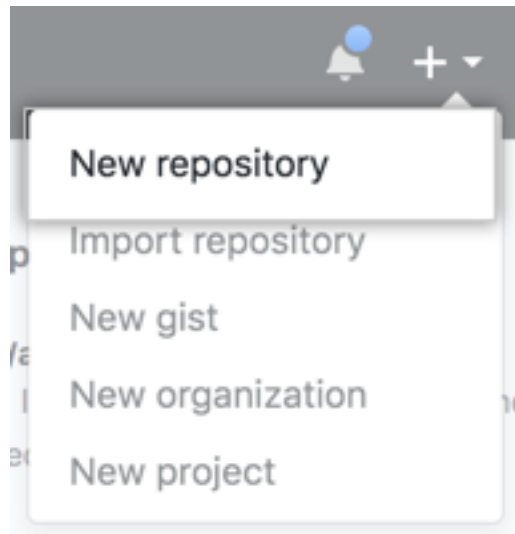
Before 2nd submission:

- Sync fork → Update branch

The screenshot displays a GitHub repository interface. At the top, the repository name 'zsli-cuhk' is visible, along with branch and tag information. The commit history table shows three recent commits: deleting a test file, adding a print statement, and a '2nd submission' of the README.md file, which is highlighted with a red box. Below the commits, the README content is shown, featuring the title 'MMAT5394' and a section 'Submission method' with a bullet point to 'Search "MMAT5394"'. An embedded image shows a GitHub search interface where 'MMAT5394' is entered, and the search results show a single repository match, 'zsli-cuhk/MMAT5394', also highlighted with a red box. The right sidebar contains sections for 'About', 'Releases', 'Packages', 'Contributors' (listing 'zsli-cuhk' and 'LIZI0001 LI, Zishang'), and 'Languages' (showing 100% Python).

Creating a repository (private, helloworld)

A repository is usually used to organize a single project. Repositories can contain folders and files, images, videos, spreadsheets, and data sets -- anything your project needs. Often, repositories include a README file, a file with information about your project. GitHub makes it easy to add one at the same time you create your new repository. It also offers other common options such as a license file. Your hello-world repository can be a place where you store ideas, resources, or even share and discuss things with others.



Owner *

Repository name *

 octocat ▾

 /

hello-world ✓

Great repository names are short and memorable. Need inspiration? How about [ubiquitous-system](#)?

Description (optional)

My first repository

☐  **Public**

Anyone on the internet can see this repository. You choose who can commit.

☒  **Private**

You choose who can see and commit to this repository.

Initialize this repository with:

Skip this step if you're importing an existing repository.

☒ **Add a README file**

This is where you can write a long description for your project. [Learn more.](#)

☐ **Add .gitignore**

Choose which files not to track from a list of templates. [Learn more.](#)

☐ **Choose a license**

A license tells others what they can and can't do with your code. [Learn more.](#)

This will set  **main** as the default branch. Change the default name in your [settings](#).

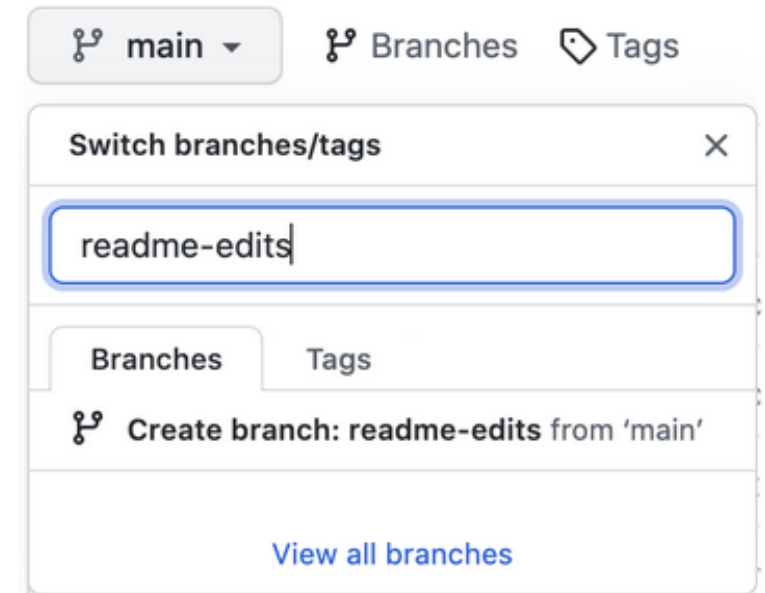
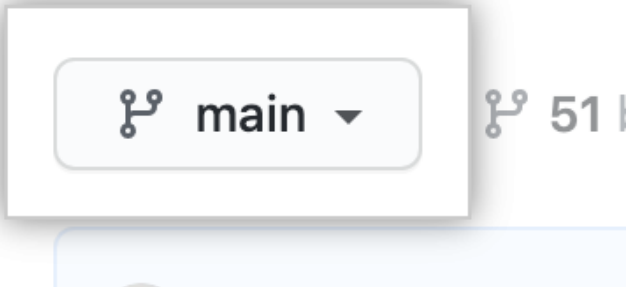
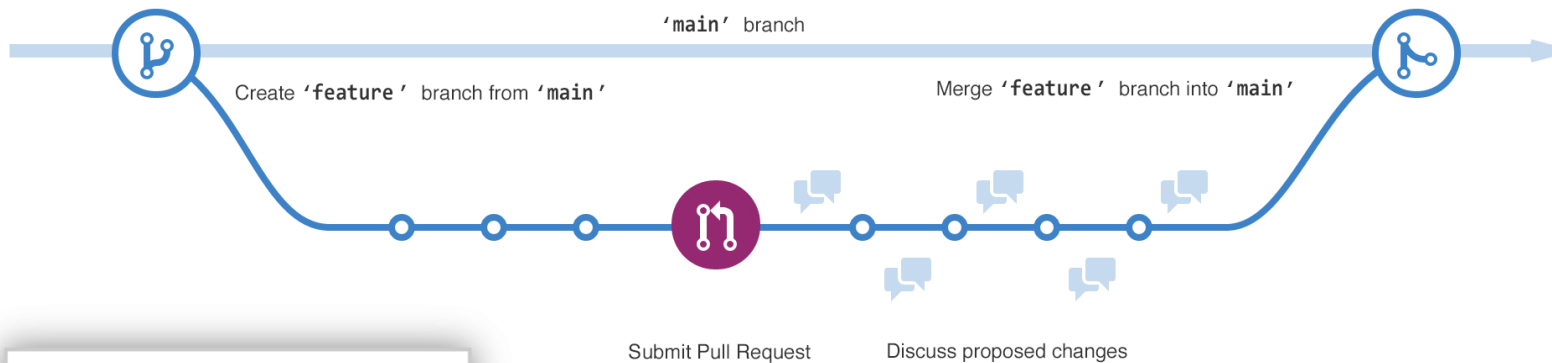
Create repository

Helloworld project

- 1) <https://github.com/>
- 2) <https://docs.github.com/en/get-started>
- 3) <https://docs.github.com/en/get-started/quickstart/hello-world>

Creating a branch

Branching lets you have different versions of a repository at one time. By default, your repository has one branch named main that is considered to be the definitive branch. You can use branches to experiment and make edits before committing them to main. When you create a branch off the main branch, you're making a copy, or snapshot, of main as it was at that point in time. If someone else made changes to the main branch while you were working on your branch, you could pull in those updates.



→ ↻ https://bitbucket.org/product/guides/basics/four-starting-steps#step-1-put-your-code-in-bitbucket

ATLASSIAN
Bitbucket Why Bitbucket ▾ Product Guide ▾ Self-Hosted Pricing Log in [Get it free](#)

← Guides home

Guide 1: A brief overview of Bitbucket

Guide 2: Getting started with Bitbucket

Meet the Bitbucket interface

Better code with Bitbucket: 4 starting steps

- **Step 1: Put your code in Bitbucket**

Step 2: Set up locally, collaborate globally

Step 3: Basic branching with Bitbucket

Step 4: Review code changes with pull request

Better code with Bitbucket: 4 starting steps

Before you begin, you'll need a [Bitbucket Cloud account](#) to follow along in this tutorial yourself. This guide will give you a tour of the parts of Bitbucket you'll use a lot, so you should also have Git installed and be familiar with basic Git commands and how Git works.

New to Git? No worries, we got you covered. [Read our Getting Git Right tutorials for a primer on Git.](#)

Now, let's put some bits in your bucket.

Step 1: Put your code in Bitbucket

Repositories (affectionately referred to as 'repos' in the biz) are where code lives in Bitbucket. There

Step 2: Set up locally, collaborate globally

Step 3: Basic branching with Bitbucket

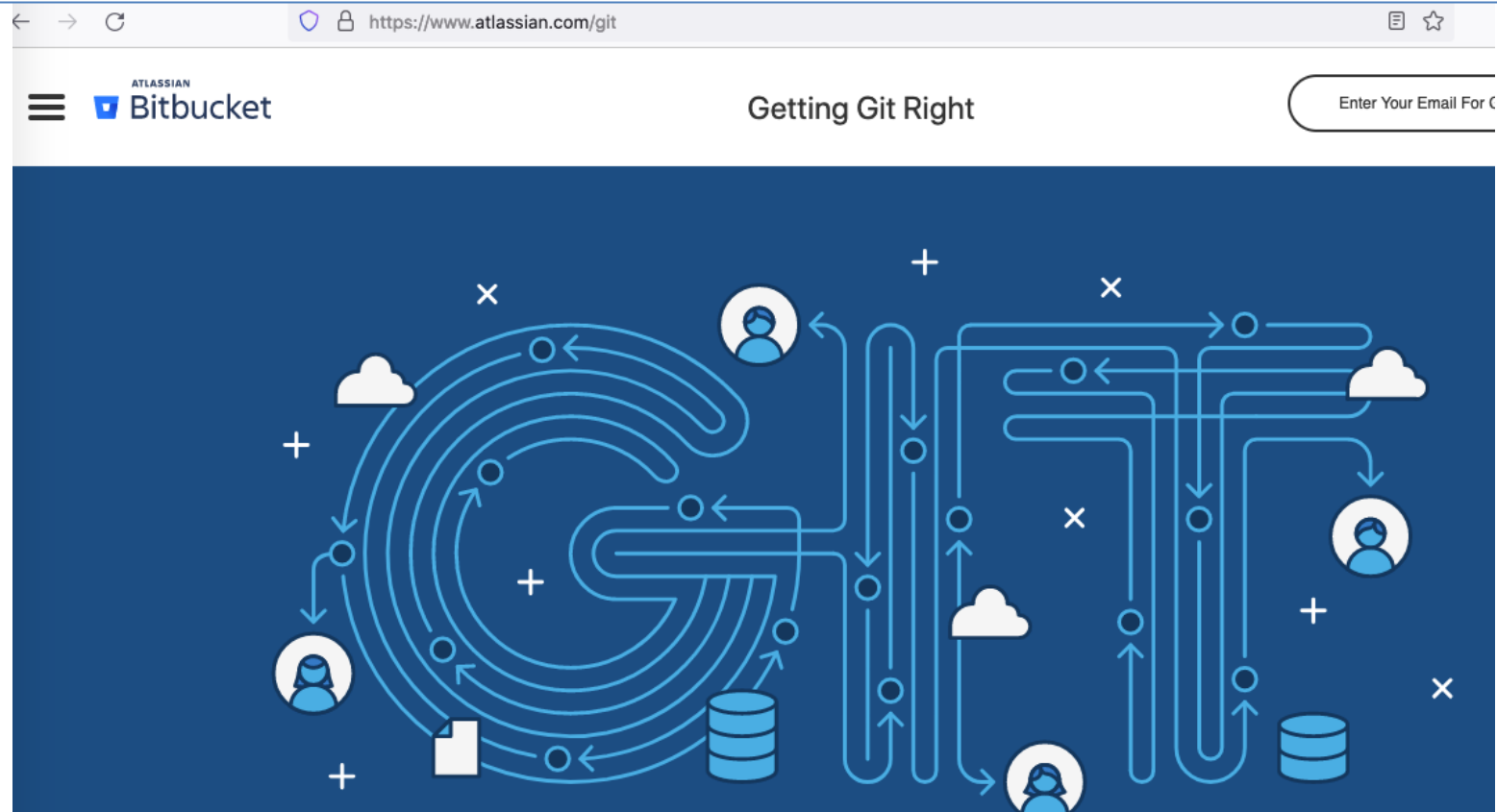
Step 4: Review code changes with a pull request

How do I create a free Bitbucket account?

Open <https://bitbucket.org/account/signup/> in your browser. Complete the fields on the form. Press Sign up.

...

1. Click a service under the You can also sign up with heading.
2. Select an account.
3. Follow the provider's prompts to complete the sign up process.
4. When prompted by Bitbucket to authenticate, choose Set a password.



GIT, GITHUB

Git is a local version control software, while GitHub is a cloud-based hosting platform built to share and collaborate on Git projects. Simply put, Git is the actual tool used to track code changes, and GitHub is a website where those tracked projects are uploaded to share with others

How Git works : Basic Overview

- 1) Create a "repository" (project) with a git hosting tool (like Bitbucket, github repository)
- 2) Copy (or clone) the repository to your local machine
- 3) Add a file to your local repo and "commit" (save) the changes
- 4) "Push" your changes to your main branch
- 5) Make a change to your file with a git hosting tool and commit
- 6) "Pull" the changes to your local machine
- 7) Create a "branch" (version), make a change, commit the change
- 8) Open a "pull request" (propose changes to the main branch)
- 9) "Merge" your branch to the main branch

GUI Clients

Git comes with built-in GUI tools for committing (**git-gui**) and browsing (**gitk**), but there are several third-party tools for users looking for platform-specific experience.

If you want to add another GUI tool to this list, just **follow the instructions**.

All

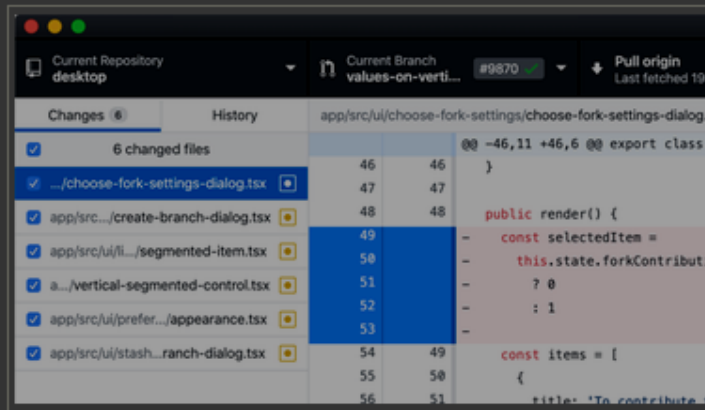
Windows

Mac

Linux

Android

iOS

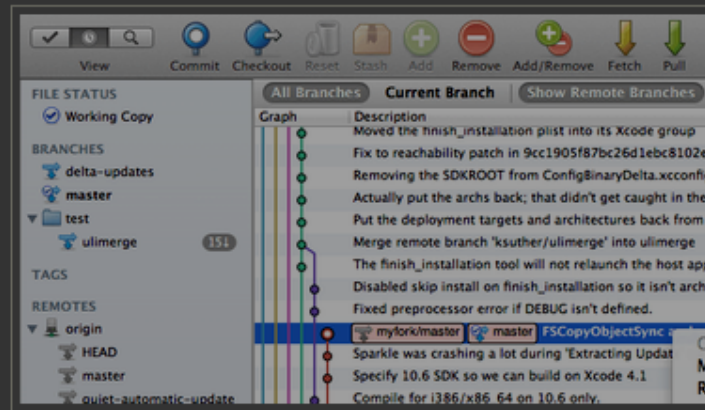


GitHub Desktop

Platforms: Mac, Windows

Price: Free

License: MIT



SourceTree

Platforms: Mac, Windows

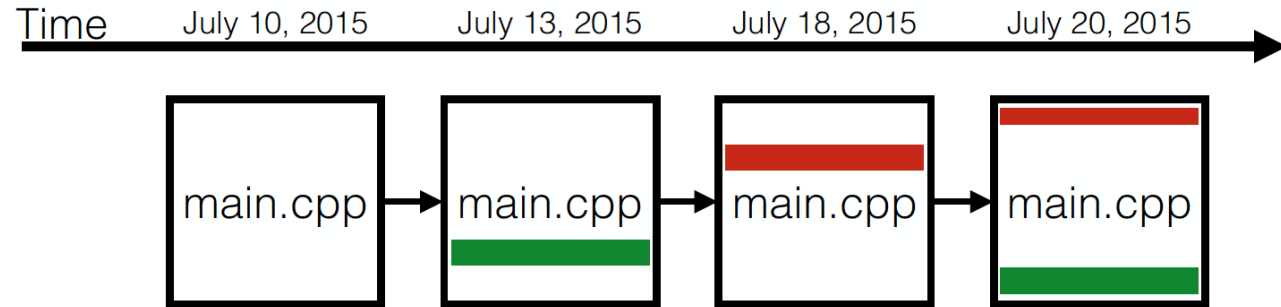
Price: Free

License: Proprietary



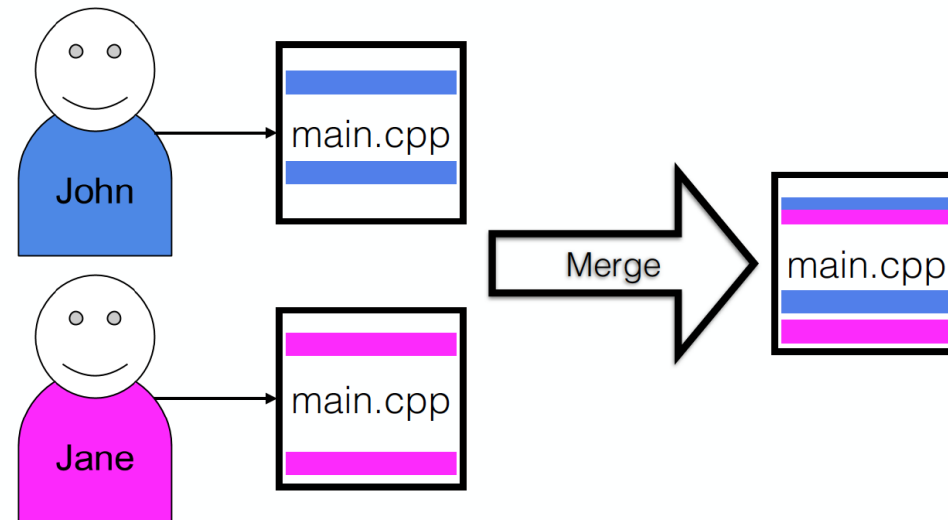
Why Use Version Control?

Tracking Changes

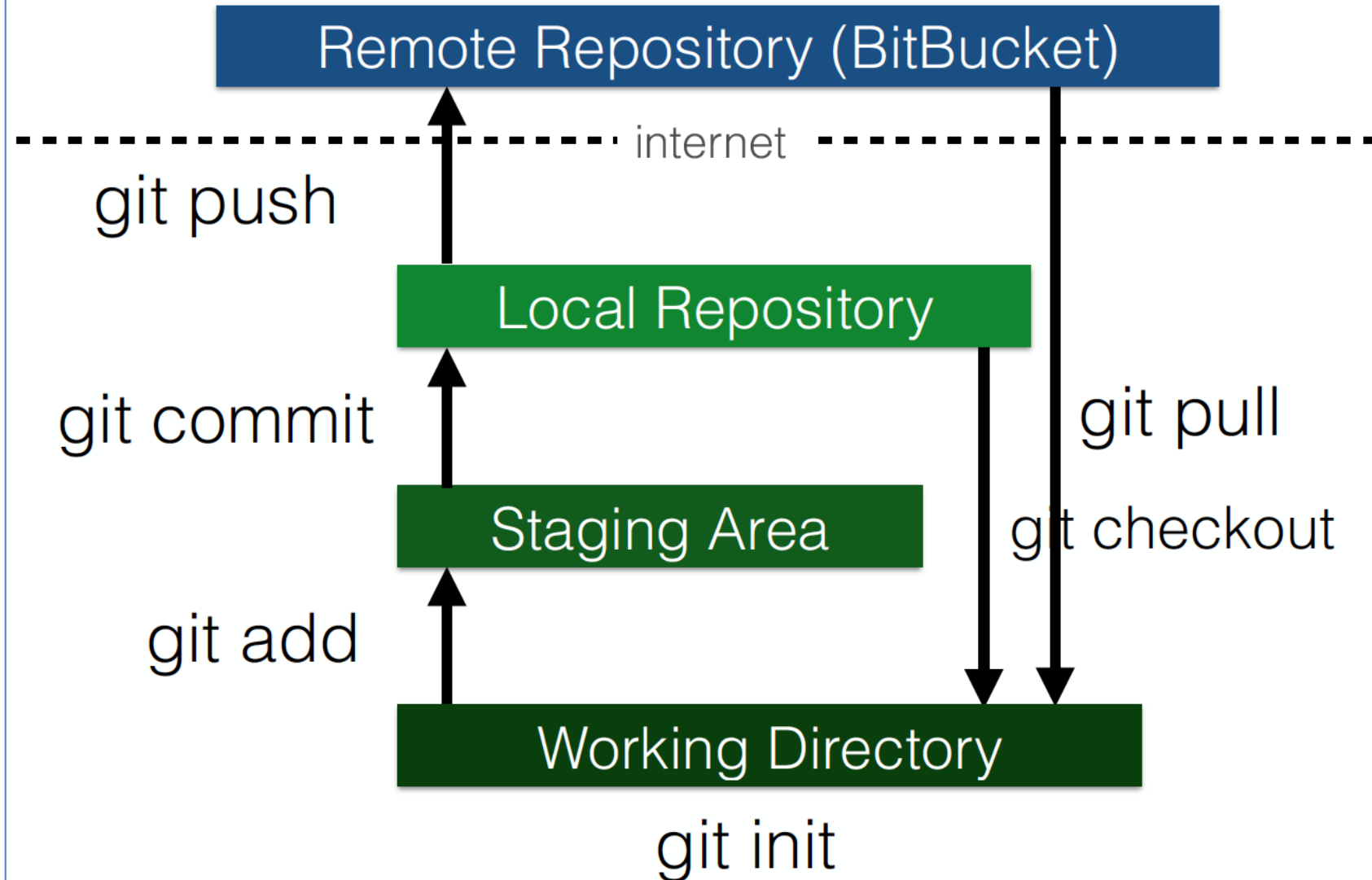


Why Use Version Control?

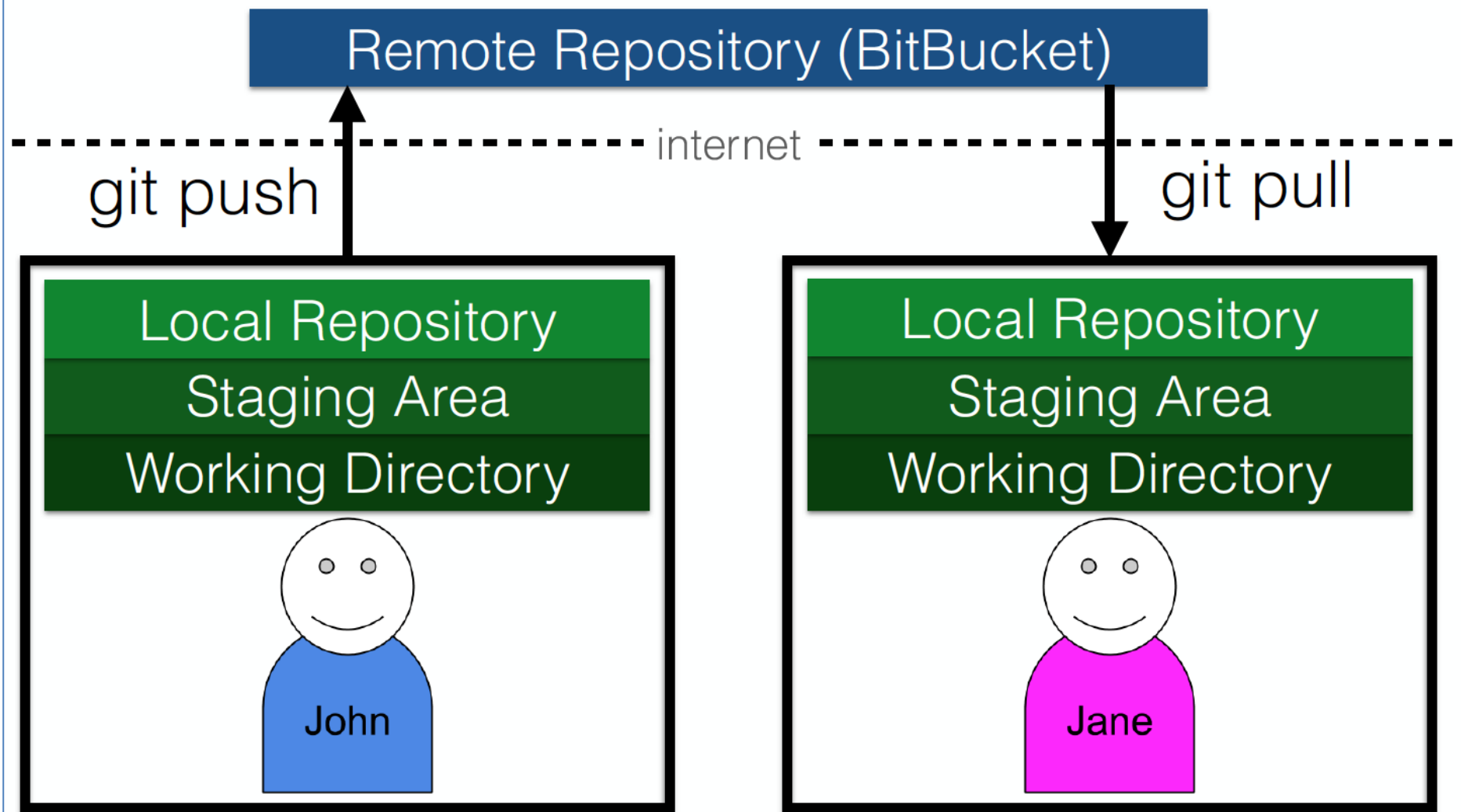
Collaboration



Git Workflow



Git Workflow



The End

Markov Chains (MC) : Software Tools

Thank You
Questions

